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Identification and predication of network attack patterns in software-defined networking

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Abstract
Software-defined networking (SDN) is earning popularity in enterprise network for simplifying network management service and reducing operational cost. However, security enhancement is required for concerns. In this paper, we analyze the network attack patterns of governments and enterprises, whose networking paradigm are constructed in SDN. In detail, methods of time series data mining including clustering and forecasting are proposed to discover hidden information in temporal network attack data. To start with, hierarchical clustering with modified dynamic time warping distance measure was developed to classify time series data of nine departments of China, which is aimed to identify patterns of network attack. We then explored autoregressive integrated moving average to build a model describing relationships and behavior of network attack as well as forecast the frequency of the future network attack, which is targeted to prevent extensive exposure of attack events. Experiments demonstrated that our models have the ability to distinguish the complex phenomena of temporal network attack and realize statistically accurate predication of network attack under SDN architecture. Our work provides the foundation for decision-making when dealing with issues of network safety.

Keywords Software-defined networking · Network attack · Statistical difference · Dynamic time warping · Autoregressive integrated moving average model

1 Introduction

Software-defined networking is an emerging network infrastructure that separates the network's control plane from the underlying data plane of network switches [1–4]. Under this situation, network switches are implemented as sample forwarding devices and the control plane is moved to a centralized application called network controller, which is in charge of entire network by simplifying policy enforcement, network configuration and so on. Therefore, the use of software-defined networking realize flexible management, well-defined programming. However, this also brings us the new challenges of potential risks, especially on network security [5, 6]. We are motivated to explore the temporal network attack in software-defined networking.

In this paper, we focus on surveying hidden worthwhile information of SDN-based temporal network attack from different government departments and enterprises. The exposure of real structure of these data would be helpful for us to maintain network safety. To this end, we implemented hierarchical clustering and forecasting for identification of attack patterns and prevention of extensive attack exposure, respectively.

To better understanding the patterns of network attack of nine departments, we first tried to hierarchically group these time series into clusters. It requires us to measure the difference between pairwise time series of network attack [7–9]. Selecting a suitable measure for evaluating similarities within data is critical because of possible time difference among time series. Here a variation of dynamic time warping (DTW) [10] is employed for dealing with timing differences and calculating time-normalized distance between pairwise time series. DTW algorithm was proposed by Hiroaki Sakoe and Seibi Chiba for spoken word recognition. It is introduced as a solution to

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overcome the weakness of sensitivity to deformation in time axis [11]. The principal of DTW is to stretch or compress them locally for eliminating the effect of highly complicated fluctuation as well as unnormalized time, which makes one resemble the other as much as possible. It involves in clustering and classification in various domains such as electro-cardiogram analysis [12–14], clustering of gene expression profiles [15, 16], signature recognition [17, 18] and vision sensors [19]. Furthermore, variations of DTW have been proposed for better choosing the possible routes of the warping paths and avoid degeneration problems, as well as reducing the time complexity of DTW calculations.

We are then motivated to employ autoregressive integrated moving average (ARIMA) [20] for modeling and predicating the frequency of network attack of each department. ARIMA model is one of the extensively used stochastic time series models [20–23]. The popularity of this model is that it flexibly characterizes several varieties of time series with simplicity and is also related with Box-Jenkins methodology [20, 21, 23, 24] for optimal model building process. The utility of ARIMA model allows us to make forecasts of network attack and provide support for making decisions.

This paper is organized as follows. In section 2, we briefly review DTW distance measure and a modification of DTW. The ARIMA model is also introduced, which has been widely applied for time series analysis. Section 3 reports hierarchical clustering with the modified DTW measure and forecasting by ARIMA model. Section 4 conducts experiments for identifying patterns of SDN-based network attack and predicating the future network attack trends. Finally, section 5 concludes the paper.

2 Background

2.1 Dynamic time warping

The goal of dynamic time warping (DTW) [10] is to find the best mapping path for comparing two time series by stretching or compressing them locally. Assume that these two time series are denoted as $X = (x_1, x_2, \cdots, x_N)$ and $Y = (y_1, y_2, \cdots, y_M)$. To be clear, the symbol $i = 1, \cdots, N$ is used to index the elements in X and $j = 1, \cdots, M$ for those in Y. As a measure of the difference between any pair of elements $x_i$and y_j, we also assume that a nonnegative, local dissimilarity function f is defined,

$$d(i,j) = f\left(x_i, y_j\right) \geq 0 \tag{1}$$

The mapping path is described by a sequence of points $p = (p_1, \cdots, p_l, \cdots, p_L)$, $p_l = (n_l, m_l)$, where $n_l \in \{1, \cdots, N\}$, $m_l \in \{1, \cdots, M\}$. This sequence can be considered to represent a warping function, which remaps the time axis from time series X to time series Y. If there is no time difference between these time series, $p_l = (l, l)$. The weighted summation of distances on warping path p is

$$D_p(X, Y) = \frac{\sum_{l=1}^{L} d(n_l, m_l) w(l)}{\sum_{l=1}^{L} w(l)} \tag{2}$$

where $w(l)$ is a per-step nonnegative weighting coefficient. $\sum_{l=1}^{L} w(l)$ is used to normalize this distance in attempt to guarantee the comparability along different paths.

To ensure reasonable warps, these conditions should be satisfied as the following restrictions on a warping path.

1) Boundary condition: $p_1 = (1, 1)$ and $p_L = (N, M)$.
2) Monotonic condition: $n_1 \leq \cdots \leq n_l \leq \cdots \leq n_L$and $m_1 \leq \cdots \leq m_l \leq \cdots m_L$.
3) Continuity condition: $n_l - n_{l-1} \leq 1$ and $m_l - m_{l-1} \leq 1$.
4) Adjustment window condition: $|n_l - m_l| \leq r$.

where r is an appropriate positive integer. Note that for any given point $p_l = (n_l, m_l)$ in the path, the possible previous neighbor is restricted to one of the three points, $(n_l - 1, m_l)$, $(n_l, m_l - 1)$ and $(n_l - 1, m_l - 1)$ (see Fig. 1).

A warping path $p = (p_1, \cdots, p_l, \cdots, p_L)$ defines an alignment between X and Y by arranging the element x_{n_l} of X to the element y_{m_l} of Y. The idea underlying DTW is to find an optimal alignment corresponding to the best path among all possible warping paths. So, the DTW distance $\mathrm{D}(X, Y)$ between X and Y is defined as

$$\mathrm{D}(X, Y) = \min_p \mathrm{D}_p(X, Y). \tag{3}$$

However, the disadvantage of the adjustment window condition is that a single element of one time series probably corresponds to many consecutive elements of the other time series. One promising solution is to constrain the slope of the acceptable warping paths. Therefore, strategies of various slope constraint conditions have been proposed in both symmetric and asymmetric forms. We here introduce one of these algorithms that has been validated to outperforms other alternatives [10].

Specifically, the recursion of the unnormalized distance is defined as follows.

$$g(n_l, m_l) = \min_{(n_l, m_l)} g(n_{l-1}, m_{l-1}) + d(n_l, m_l) w(l) \tag{4}$$

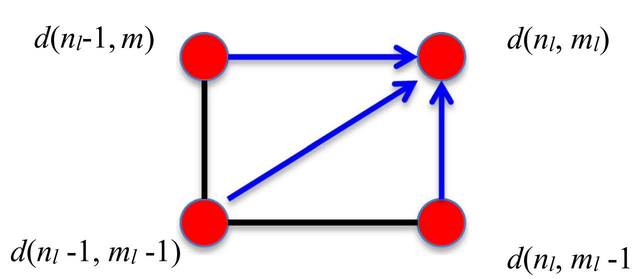


Fig. 1 Local constraint for searching the best mapping path

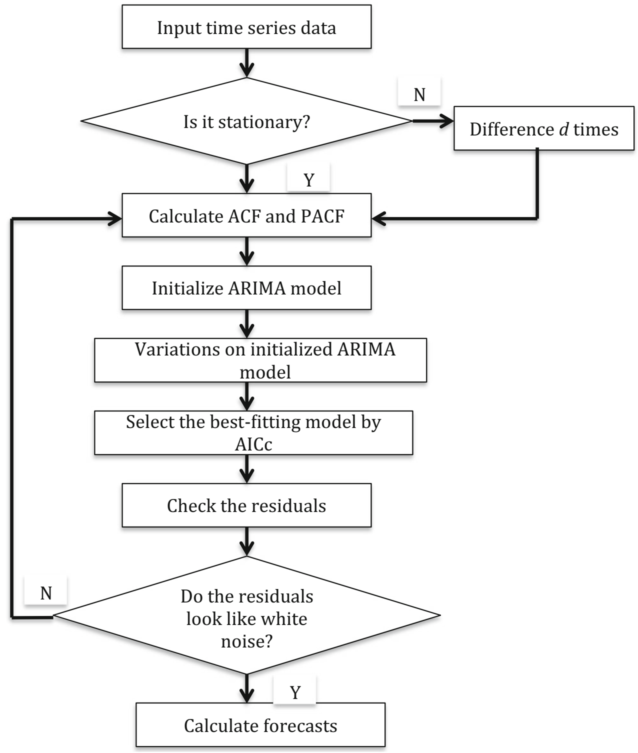


Fig. 2 Process for forecasting by an ARIMA model

And, the DTW distance D(X, Y) between X and Y becomes,

$$\mathrm{D}(X,Y) = \frac{1}{N} g(n_L, m_L) \tag{5}$$

where $N = \sum_{l=1}^{L} w(l)$.

Then the modification of DTW is defined,

$$g(n,m) = min \begin{cases} g(n-1,m-1) + 2d(n,m) \\ g(n-2,m-1) + 2d(n-1,m) + d(n,m) \\ g(n-1,m-2) + 2d(n,m-1) + d(n,m) \end{cases} \tag{6}$$

where $(n,m) \in [1,N] \times [1,M] \backslash \{(1,1)\}$. The initial values are set to $D(1,1) = c(x_1, y_1)$, $D(n,-2) = D(n,-1) = D(n,0) = \infty, n \in [-2,N]$; $D(-2,m) = D(-1,m) = D(0,m) = \infty, m \in [-2,M]$. The slopes of the generating warping paths range from 1/3 to 3.

2.2 ARIMA model

In order to capture the underlying structure of the time series data from nine departments, the Autoregressive Integrated Moving Average (ARIMA) is proposed for time series analysis. This model-building process, introduced by Box and Jenkins, is designed to be a combination of Autoregressive (AR) and Moving Average (MA) models. In this model, the current observation of the series is a linear function of several past observations and random errors. Usually, Combining AR(p) and MA(q) models, the ARIMA model is a form of ARIMA(p, d, q), where p is the order of Autoregressive term, q is the order of the moving average term and d is the order of the differencing.

Assume that observations y_t are recorded at times $t = 1, 2, \ldots, n$. ARIMA(p, d, q) can be written as

$$\varphi(L) \nabla^d y_t = \theta(L) \varepsilon_t \tag{7}$$

where ε_t is the random error (or random shock) at time period t. φ(L) and θ(L) are autoregressive operator of order p and moving average operator of order q. To be specific, $\varphi(L) = 1 - \varphi_1 L - \varphi_2 L^2 - \cdots - \varphi_p L^p$, $\theta(L) = \theta_1 B - \theta_2 B^2 - \cdots \theta_q B^q$; φ_i, $i = 1, \cdots, p$ and θ_j, $j = 1, \cdots, q$ are parameters. L and $\nabla = 1 - L$ are backward shift operators that can be defined as $L^m y_t = y_{t-m}$ and $\nabla y_t = y_t - y_{t-1}$ with $\nabla^d = \nabla \nabla^{d-1}$. The random shock ε_t is assumed to be a white noise process, $\varepsilon_t \sim \mathrm{WN}(0, \sigma_\varepsilon^2)$.

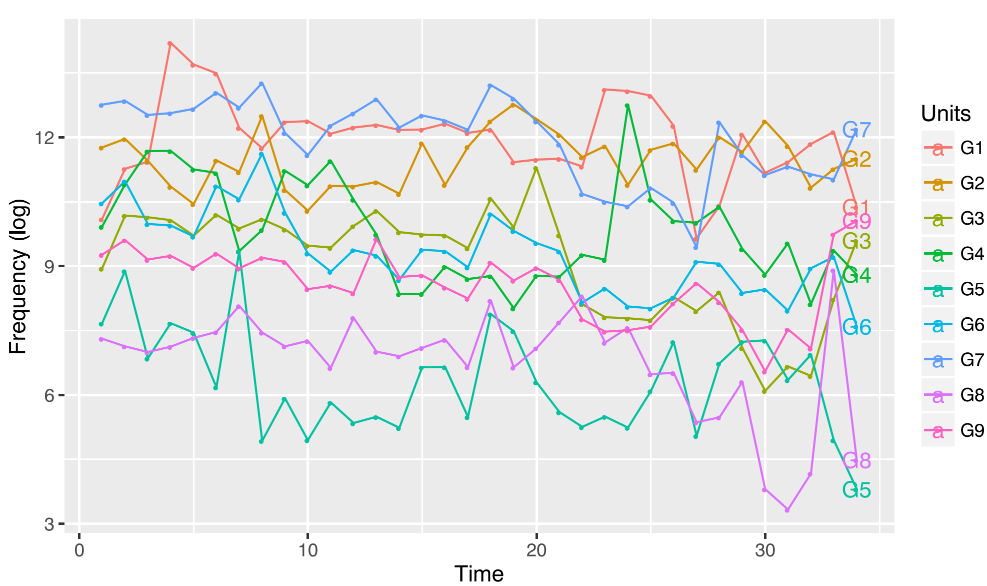


Fig. 3 Weekly frequency of Network attack in SDN. Observed frequency of important network is displayed for departments from G1 to G9

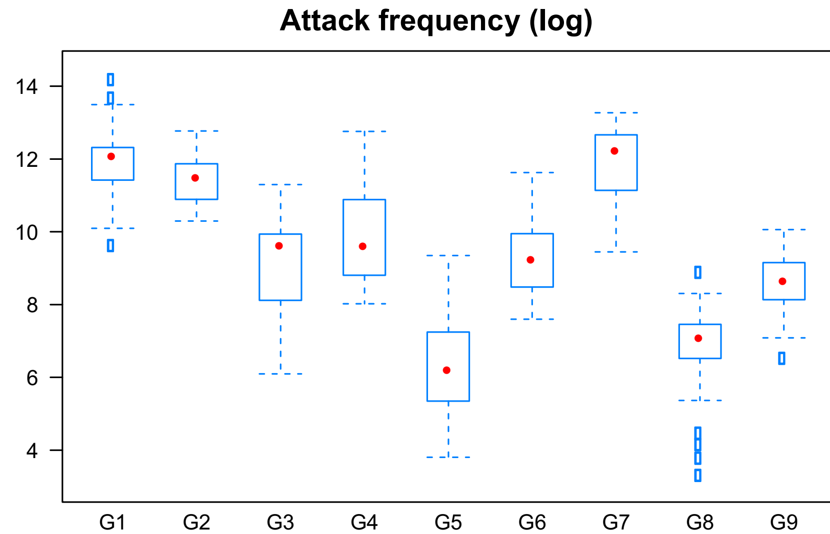


Fig. 4 Statistical distributions of groups of observations for nine departments

3 Analytic approach

3.1 Time series clustering using DTW distance measure

The target of clustering time series is exploring natural groups for the datasets [7–9] and mining the valuable information. We usually acquire clusters by maximizing the dissimilarity between clusters and minimizing the dissimilarity within clusters. Especially, the choice of appropriate distance measure is a critical step to characterize the natural groups of time series data during clustering. One of the classic measure methods is dynamic time warping, which calculates pairwise distance based on the shape of the time series. The rationale behind DTW is to calculate the optimal match between two times series subject to certain restrictions. It aligns two time series in the time dimension using the shortest non-linear warping path.

Algorithm: Optimal warping path

Input: Two time series data X and Y.

Output: Optimal warping path $p*$.

Procedure:

1. Define $\mathrm{D}(n, m)$ as the DTW distance between $(x_1, x_2, \cdots, x_n)$ and $(y_1, y_2, \cdots, y_m)$ with the mapping path starting from $(1,1)$ to (n, m), where $n \in \{1, \cdots, N\}, \mathrm{m} \in \{1, \cdots, M\}$.

2. The optimal path $p^* = (p_1, \cdots, p_\mathrm{L})$ is computed in reverse order of the indices starting with $p_L = (N, M)$. Suppose $p_\ell = (n, m)$ has been computed. In case $(n, m) = (1,1)$, one must have $\ell = 1$ and we are finished. Otherwise, $p_{\ell-1}$is calculated by the variation of DTW, which is defined in (6).

3. Get the optimal path: $p^* = (p_1, \cdots, p_L)$.

Table 1 Multiple comparison test between groups of network attack after Kruskal-Wallis test

Comparison	Obs.dif	Difference	Comparison	Obs.dif	Difference
G1-G2	17.000	F	G3-G7	123.147	T
G1-G3	120.294	T	G3-G8	87.676	T
G1-G4	90.264	T	G3-G9	26.558	F
G1-G5	218.411	T	G4-G5	128.147	T
G1-G6	113.705	T	G4-G6	23.441	F
G1-G7	2.852	F	G4-G7	93.117	T
G1-G8	207.970	T	G4-G8	117.705	T
G1-G9	146.852	T	G4-G9	56.588	F
G2-G3	103.294	T	G5-G6	104.705	T
G2-G4	73.264	T	G5-G7	221.264	T
G2-G5	201.411	T	G5-G8	10.441	F
G2-G6	96.705	T	G5-G9	71.558	T
G2-G7	19.852	F	G6-G7	116.558	T
G2-G8	190.970	T	G6-G8	94.264	T
G2-G9	129.852	T	G6-G9	33.147	F
G3-G4	30.029	F	G7-G8	210.823	T
G3-G5	98.117	T	G7-G9	149.705	T
G3-G6	6.588	F	G8-G9	61.117	F

Employing the variation of DTW as distance measure, hierarchical clustering seeks to build a hierarchy of clusters and merge similar clusters. It provides us a distinctive pattern portrait of network attack, and that these patterns could be classified into subtypes. Therefore, attack patterns of nine departments are analyzed by hierarchical clustering. The corresponding results may present distinct subtypes of network attack, which would help us identify attack types.

3.2 Predication by ARIMA model

Having demonstrated the choice of ARIMA process for modeling time series data, it is necessary to determine the values of parameters p, d and q. However, there is no statistical theory for determining the parameters p, d and q. Instead, these parameters can only be estimated by heuristic methods. To do this, we start with employing unit root test to find a proper value d_0 for parameter d. Rather than considering every possible combination of p and q, two steps are executed to select parameters p and q. In the first step, ACF and PACF analysis is carried out to initialize the parameters p, q as p_0, q_0 after differencing the data d_0 times. In the second step, we vary p and q by ± 1 from the initial model ARIMA (p_0,d_0,q_0) and obtain a group of extended ARIMA models, ARIMA($p_0 \pm 1$,d_0,$q_0 \pm 1$). Lastly, the best-fitting model is selected from these models by minimizing the AICc.

The best-fitting model built for each department reveals its trend of frequency of network attack. To test the performance of these models, a portmanteau test is executed to investigate whether the residuals of our model are white noise. Moreover, these models can be applied for predicating the future values of attack for nine departments. The procedures of forecasting by ARIMA models are shown in Fig. 2.

4 Experimental analysis

4.1 Datasets

We investigate network attacks of government departments and enterprises, for which the networks are built in SDN architecture. It heavily occurs that varieties of temporal attack events are performed by individuals from different locations. More precisely, we record the important attacks, including sql injection, file download, cross-site scripting, application vulnerability exploit, webshell and file parsing exploits. Here, our work focuses on the weekly important network attacks of nine departments in attempt to prevent the extensive exposure of privacy information and report the overall frequency of the important attacks of each department for 34 weeks. Therefore, it provides us overall 30×9 data points, in which each column corresponds to attack times series of a department. Considering the drastically local

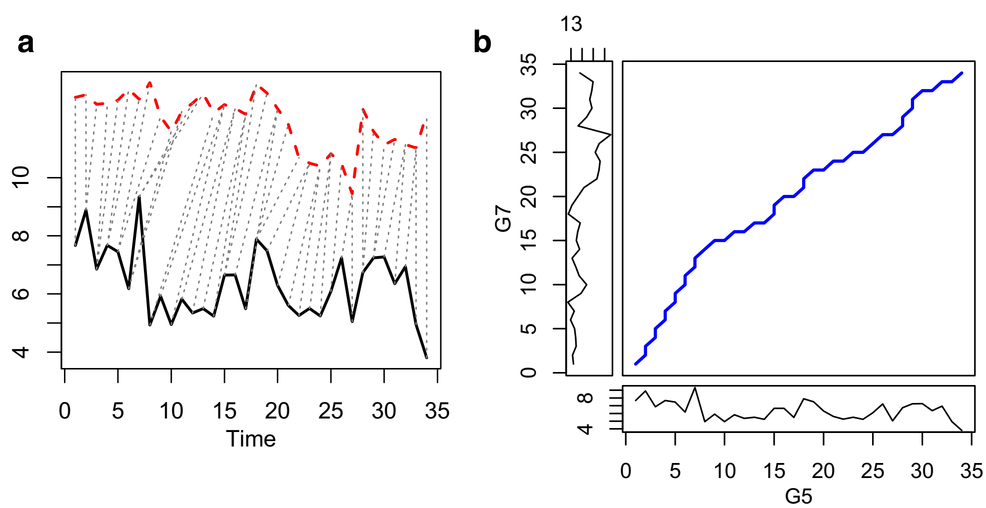


Fig. 5 Dynamic time warping results of departments G5 and G7. (**a**) Optimal time alignment of two time series; (**b**) Optimal warping path

Table 2 Distance calculations between time series using DTW distance measure

	G1	G2	G3	G4	G5	G6	G7	G8	G9
G1	0	17.696	64.840	43.709	156.349	59.479	14.738	140.932	82.227
G2	17.696	0	54.277	34.228	143.516	48.331	16.592	129.707	69.613
G3	64.840	54.277	0	32.250	88.160	16.758	65.618	69.225	18.293
G4	43.709	34.228	32.250	0	98.154	20.422	43.498	84.899	33.282
G5	156.349	143.516	88.160	98.154	0	90.376	157.288	37.619	69.000
G6	59.479	48.331	16.758	20.422	90.376	0	59.121	76.168	21.129
G7	14.738	16.592	65.618	43.498	157.288	59.121	0	142.865	82.152
G8	140.932	129.707	69.225	84.899	37.619	76.168	142.865	0	53.476
G9	82.227	69.613	18.293	33.282	69.000	21.129	82.152	53.476	0

fluctuations of the data, we preprocess the data by log transformation. For convenience, the nine departments are simply demonstrated from G1 through G9 (see Fig. 3).

4.2 Statistical methods for comparing the level of attack frequency of departments

The variance of analysis was performed for observations of network attack of nine departments, which is beneficial to statistically compare and distinguish patterns of temporal network attack. To do this, we first conducted Shapiro-Wilk test [25] and Bartlett's test [26], allowing us to test normality of observations and the homogeneity of variance, respectively. Estimate of the p-value was then computed for these two tests, p-value = 4×10^{-4} and p-value = 0.001. As a result, these two hypotheses were rejected at level α=0.01. It indicated that the non-parametric test should be chosen to perform the variance of analysis.

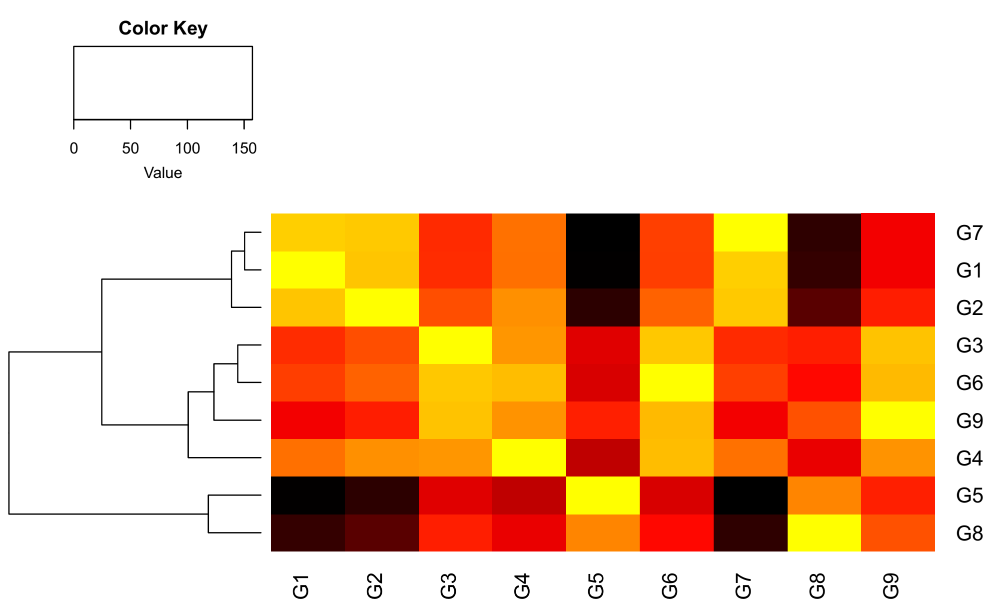
Kruskal-Wallis rank sum test [27] is a statistical approach that can be used to determine whether the distributions of different groups of data are identical without assuming them to follow the normal distribution (See Fig. 4). We applied the Kruskal-Wallis test to compare the time series data. The null hypothesis is that the weekly important frequency of network attack has the identical distribution for nine departments. It turned out that the p-value is nearly zero, p-value <0.001. Therefore, the null hypothesis was rejected. It suggested that the patterns of network attack are not completely identical for the nine departments and at least one of the groups is different from others.

From the analysis above, it is desired to see exactly which groups of attack pattern are significantly different. To this end, a multiple comparison test was performed to determine the difference between groups with pairwise adjusted comparisons appropriately [28]. For each pairwise comparison, the test reported the observed difference between the group means. Those pairs of groups that have observed differences higher than a critical value were considered statistically different at a given significance level. The results of pairwise comparison between groups were displayed in Table 1. It can be found that most of between-group network attacks are significantly different.

4.3 Hierarchical clustering for time series of nine departments

The performance of clustering heavily depends on a meaningful distance function to cluster data vectors that are close to

Fig. 6 Hierarchical clustering for time series of nine departments based on modified DTW distance measure



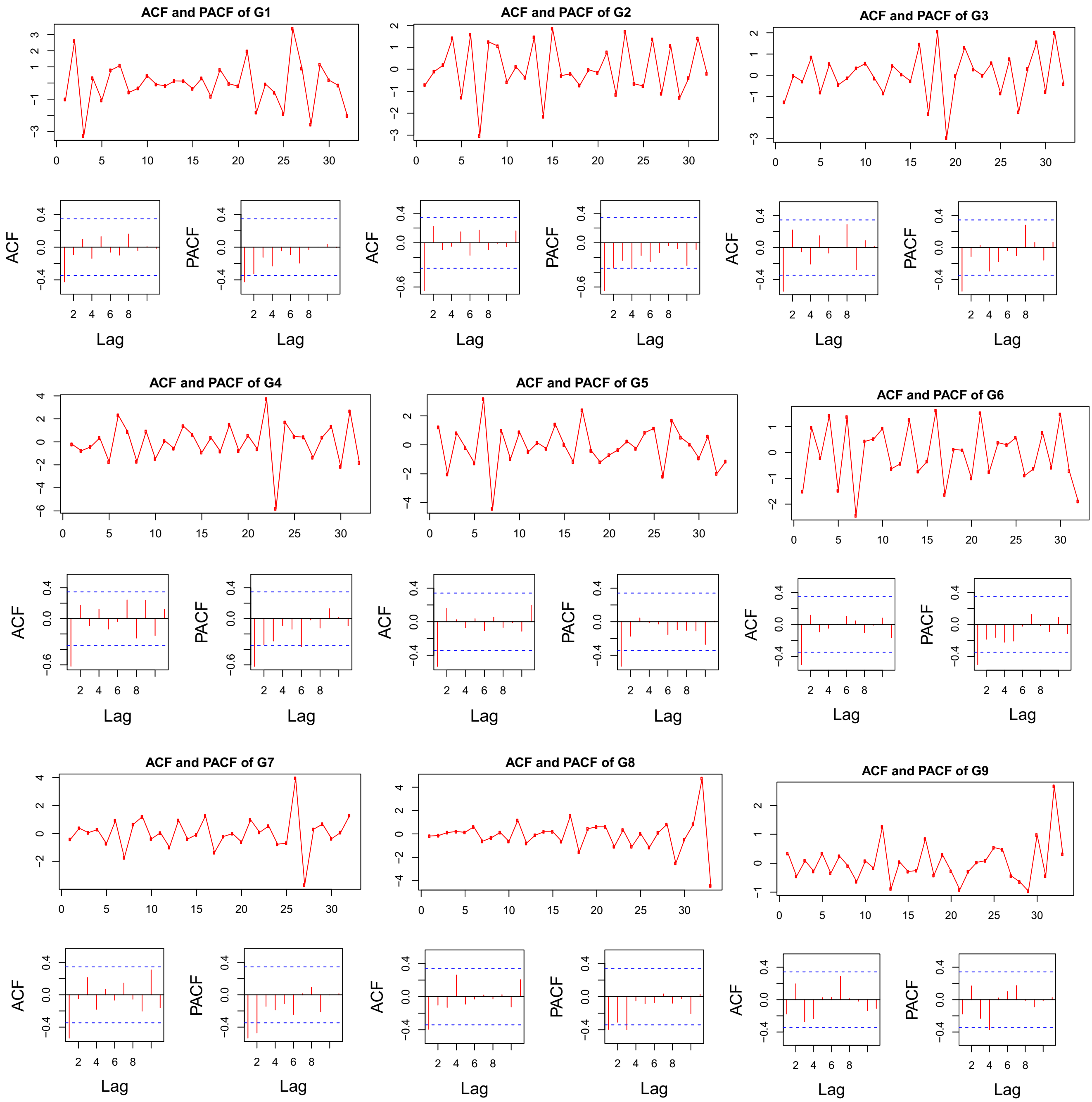


Fig. 7 Differenced time series data and the corresponding ACF and PACF for the differenced data of nine departments

each other and distinguish them from others that far away. Dynamic time warping owns the capability of comparing time series by stretching or compressing them locally in order to make one resemble the other as much as possible. To avoid degeneration problem during comparing temporal network attack, a variation of DTW is employed for determining the optimal warping path between temporal network attack sequences. To be clear, an example is given for illustration. The modified DTW distance between the two time series from departments G5 and G7 was calculated by summing the distances of individual aligned elements. Figure 5a displayed the optimal alignment between the two time series, which corresponds to the best warping path in the Fig. 5b.

Calculating the modified DTW distance for each pairwise time series (See Table 2) to measure the similarity between time series, we implemented hierarchical clustering [7, 9] for identifying network attack patterns of nine departments (see Fig. 6). From the results, the situation of network attack of nine departments is approximately distinguished into three groups. We also found that the result of the hierarchical clustering is basically consistent with that of multiple comparison tests, which demonstrated our availability for correctly classifying times series.

Table 3 Comparison of different ARIMA models for nine departments. Different ARIMA models are the combination of different values of parameters *p*, *d* and *q*

AICc	(p_0,d_0,q_0)	(p_0+1,d_0,q_0)	(p_0+1,d_0,q_0+1)	(p_0,d_0,q_0+1)	(p_0-1,d_0,q_0)	(p_0-1,d_0,q_0-1)
G1	99.67	102	104.92	104.92	97.24	112.82
G2	75.22	77.64	79.16	75.96	76.78	103.16
G3	86.86	89.01	91.75	89.55	84.48	97.84
G4	103.48	106.07	107.34	105.05	103.29	127.26
G5	112.88	115.23	116.81	115.43	111.04	111.04
G6	78.72	81.07	79.08	76.39	77.69	98.23
G7	83.22	85.98	88.8	85.77	81.99	95.98
G8	110.24	110.24	110.24	113.07	111.65	111.65
G9	111.65	111.65	111.65	111.65	73.69	_

4.4 ARIMA model for predicating network attack

To help explain the attack behavior and forecast the future values, the use of ARIMA model was implemented for modeling these weekly time series data. An ARIMA (*p*, *d*, *q*) model is represented by three parameters, *p*, *d* and *q*, where the parameter *p* is the degree of autoregressive, *d* is the degree of integration, and *q* is the degree of moving average. It requires estimating these three parameters in order to find the best-fitting ARIMA model for time series data of each department.

To do this, we first applied log transformation for preprocessing the time series (See Fig. 3). This data adjustment was aimed to suppress large fluctuations. However, it was obviously observed that these time series data from different departments are definitely non-stationary. In view of this situation, the unit testing is used to initialize the order of difference,

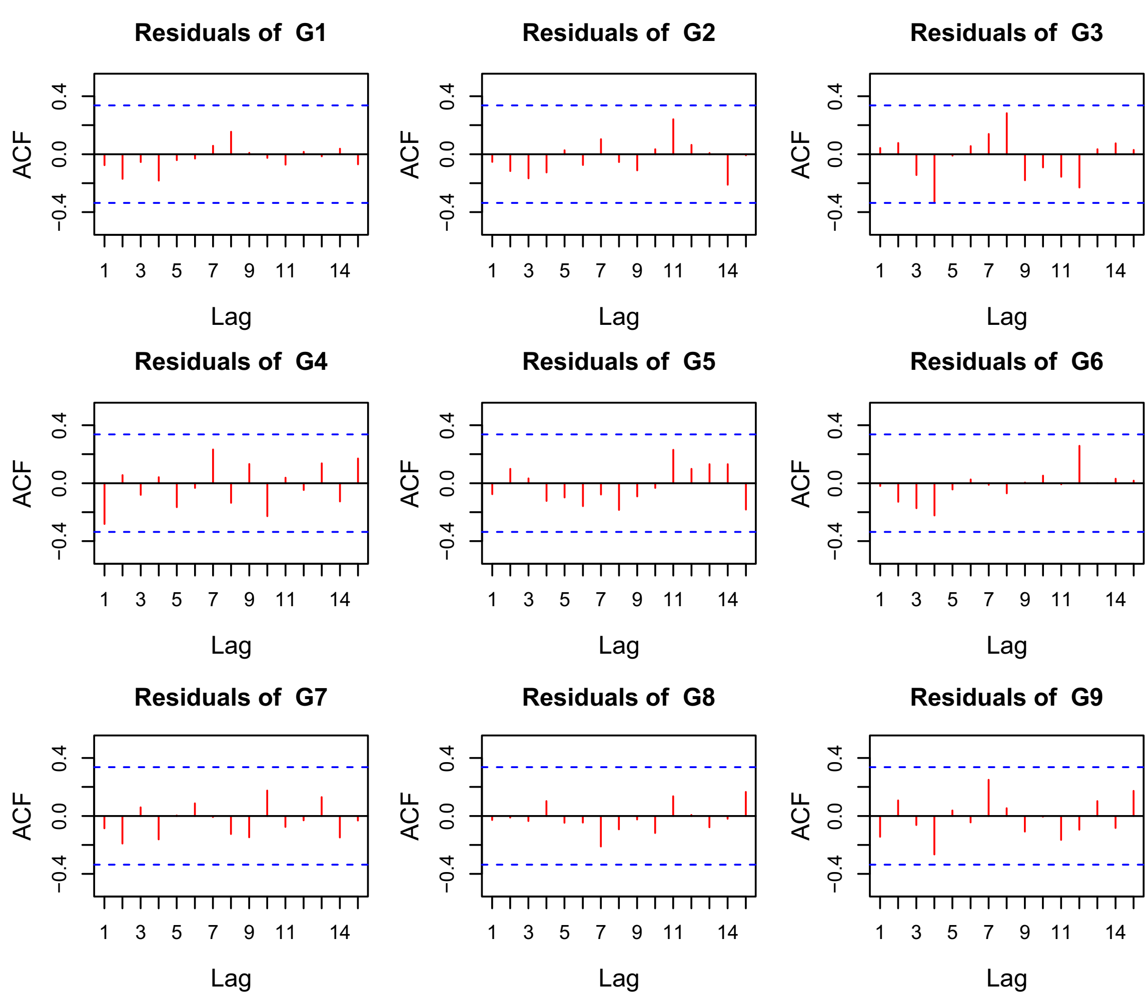


Fig. 8 Residual plots for the best-fitting ARIMA models of frequency of network attack from different departments. These plots are aimed to intuitively observe the performance of the fits

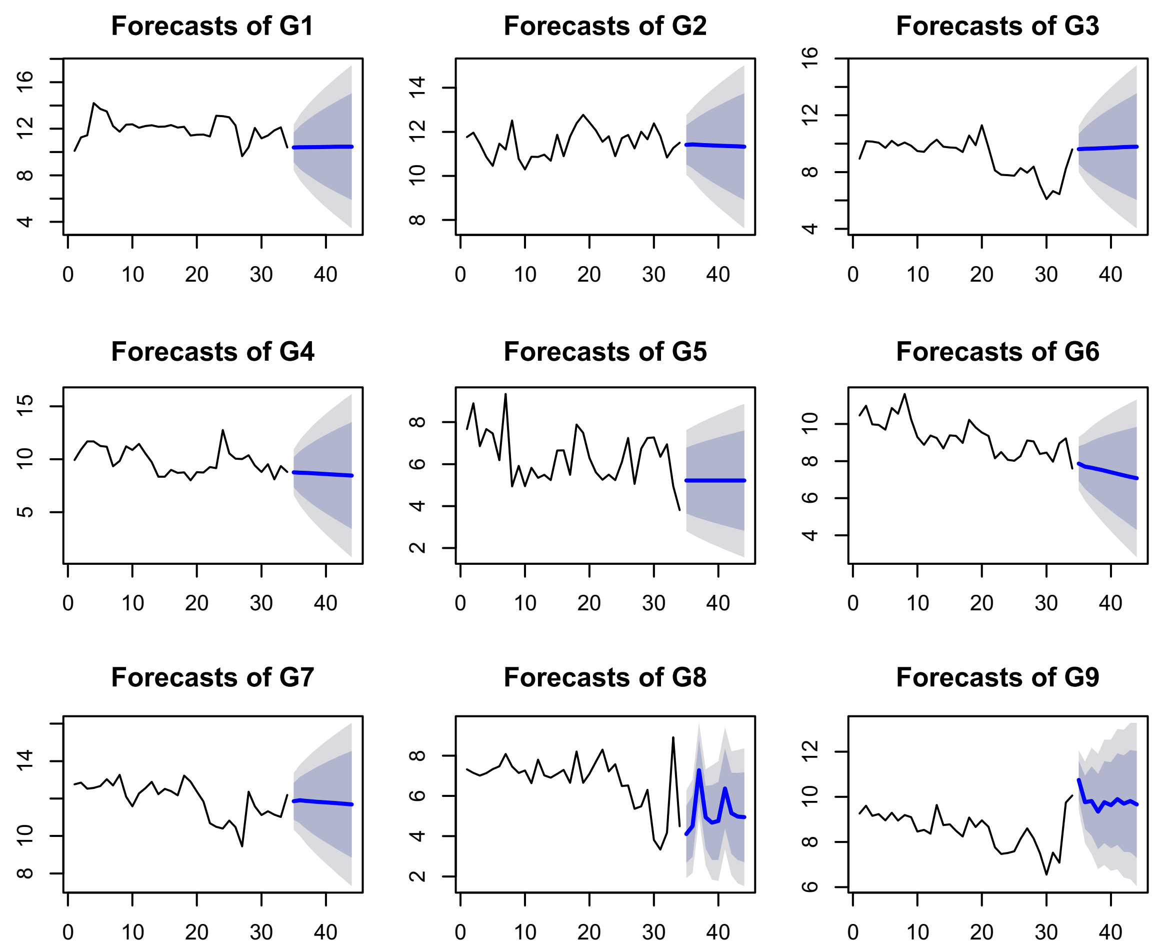


Fig. 9 Forecasts for the frequency of network attack from nine departments

which is able to guarantee its being non-stationary. We then examined the ACF and PACF, and initialized the parameters p and q (See Fig. 7). Given initialized parameters p_0, d_0 and q_0, candidate ARIMA models, ARIMA (p_0,d_0,q_0), were built for charactering network attack patterns of nine departments, individually represented as ARIMA (1,2,1), ARIMA(1,2,1), ARIMA(1,2,1), ARIMA(1,2,1), ARIMA(1,1,1), ARIMA(1,2,2), ARIMA(2,2,1), ARIMA(3,1,1), ARIMA(4,1,0).

The AICc was then employed to search for a better model for the description of each time series data. Specifically, an initialized ARIMA(p_0,d_0,q_0) model was fitted along with variations of p_0 and q_0, including $p_0 + 1$, $p_0 - 1$, $q_0 + 1$, $q_0 - 1$. Different combinations of these values correspond to different ARIMA models. We therefore acquired models including ARIMA($p_0 + 1$,d_0,q_0), ARIMA($p_0 + 1$, d_0,$q_0 + 1$), ARIMA(p_0, d_0,$q_0 + 1$), ARIMA(p_0–1,d_0,q_0). Of these, the best-fitting model is the one that corresponds to the smallest AICc value (See Table 3). These selected best models of nine departments are denoted as ARIMA(0,2,1), ARIMA(1,2,1), ARIMA(0,2,1), ARIMA(0,2,1), ARIMA(0,1,1), ARIMA(1,2,1), ARIMA(1,2,1), ARIMA(3,1,1), ARIMA(3,1,0).

Having built ARIMA models for time series data above, it is of necessity to test the performance of models. To finish this goal, we first tested the best-fitting models by plotting the ACF of residuals (See Fig. 8). The ACF test results showed all correlations within the threshold limits, which suggested that the residuals behave like white noise. Furthermore, a portmanteau test was implemented to test the performance of models. It also indicated that the residuals are white noise. Therefore, all of the best-fitting ARIMA models are appropriate for describing the behavior of network attack. Last but the most important is that these models were used to predicate the future values in attempt to control and prevent the possible extensive explosions of the network attack (See Fig. 9).

5 Conclusion

Software-defined networking (SDN) provides an opportunity for solving problems of traditional networks, which has succeeded in paving the path towards the next generation networking. We are accordingly confronted with the challenge of the network security. In this paper, methods of time series data mining have been proposed to analyze temporal network attacks collected from different government departments and enterprises, whose networking paradigm is constructed in software-defined networking. The main work has been implemented as follows.

First, we performed statistical analysis for these temporal data. Specifically, the variance of analysis was performed for comparing time series data. It demonstrated that patterns of temporal network of nine departments are significantly different. Furthermore, the multiple comparison test was performed to determine the difference between groups with pairwise adjusted appropriately.

Second, we applied clustering for hierarchically grouping network attack of nine departments. Considering distortions of time series data, DTW-based measure was proposed for hierarchical clustering. This work is helpful to identify the patterns of network attack of nine departments.

Finally, we used autoregressive integrated moving average for modeling the behavior of network attack of and predicating the frequency of the future network attack. The purpose of this work is to prevent extensive exposure of attack events.

In conclusion, analysis of network attack patterns was conducted by the use of statistical tests, hierarchical clustering based on a modified DTW distance measure and forecasting by ARIMA models. It can provide useful information for guaranteeing cybersecurity under the environment of software-defined networking.

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